Lab 2: Improving Agent Quality through AI Instructions and Prep data for AI

Overview

In this lab, you will build a Fabric Data Agent that answers questions using the ecommerce-dataset semantic model and configure it for improved responses. You'll start by creating the agent and exploring the model through a few sample prompts while inspecting the run steps (paraphrasing, DAX generation, and query output). Next, you'll add Agent Instructions to guide how the agent asks clarifying questions and formats responses. Finally, you'll use Prep data for AI to refine the AI-ready schema and verified answers so the agent selects the right tables and columns and produces more accurate, consistent results.

Prerequisites

1. You have completed Lab 1 – Getting Started

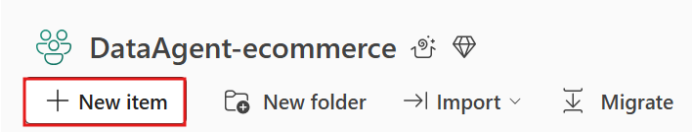
Learning Objectives

1. Create a data agent with semantic model as the data source
2. Learn how to set up Prep data for AI configuration and agent instructions

Lab Steps

Step 1: Create a Data Agent

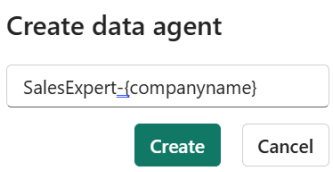
1. Select "+ New Item"



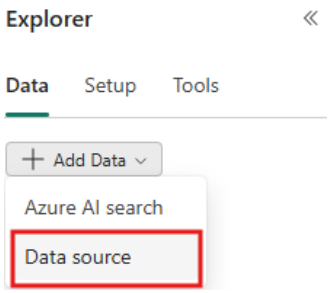
2. Search for "Data agent" and Select "Data agent"

New item
Favorites
All items
Data agent
Filter
Data agent (preview)
Build a generative AI agent that understands your data and can answer complex questions in a variety of conversational interfaces.

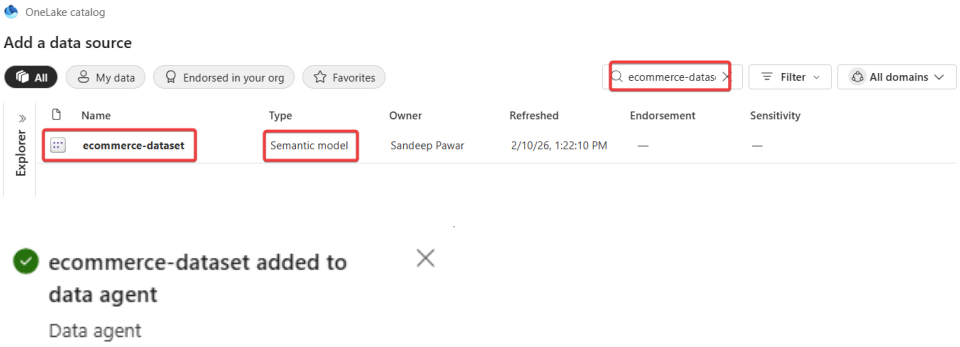
3. Name the Data Agent “SalesExpert-{WORKSPACE OR COMPANY NAME}” -> “Create”



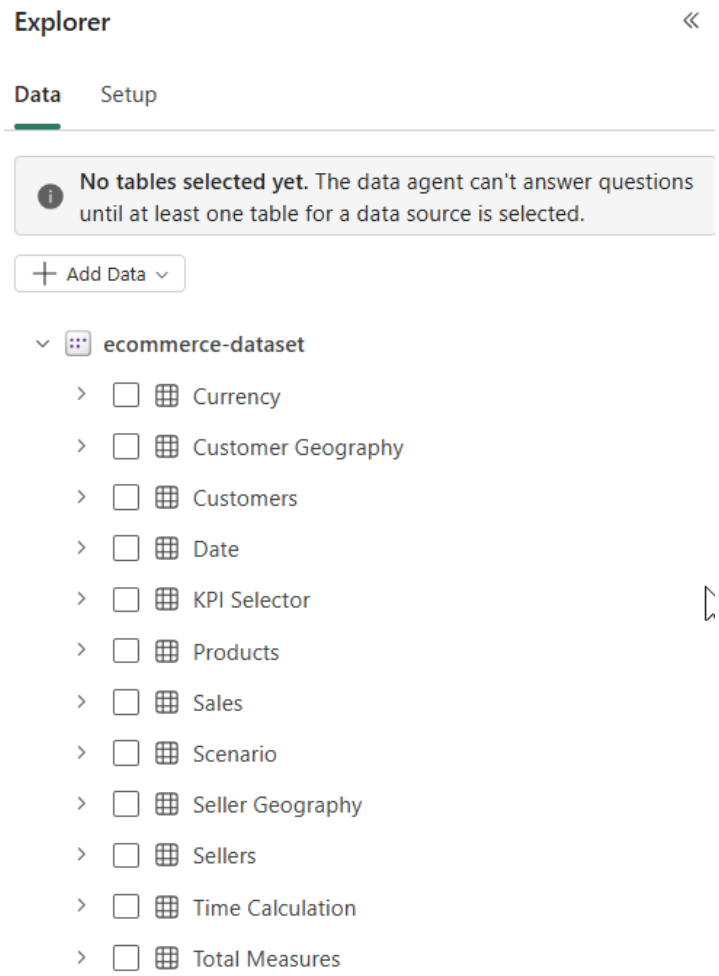
4. Click “+ Data source” in the Explorer



5. Search for “ecommerce-dataset” using its name, select the semantic model, and click “Add”



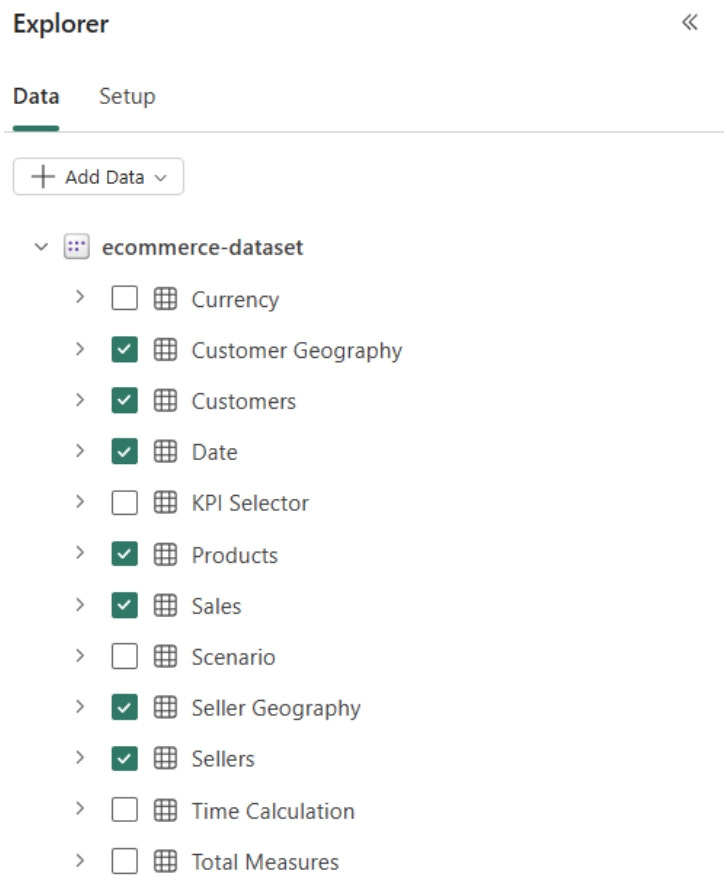
6. You should see the semantic model with all the tables in the model:



Step 2: Data exploration through Data Agent

For this lab, we will explore the data in the semantic model

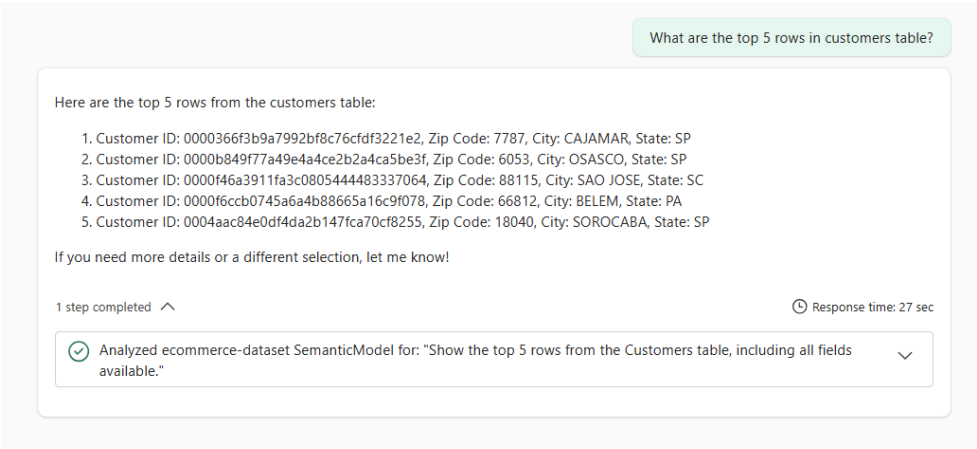
1. In the Explorer pane, select the following tables. The unselected tables will be out of scope of the data agent:


Explorer
Data
Setup
Add Data
ecommerce-dataset
Currency
Customer Geography
Customers
Date
KPI Selector
Products
Sales
Scenario
Seller Geography
Sellers
Time Calculation
Total Measures

2. Before we configure the data agent further, let's start by asking a few questions.

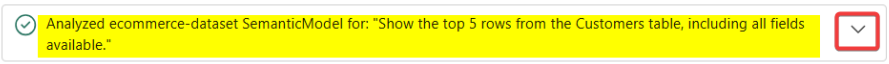
   Ask the Data Agent - **What are the top 5 rows in customers table?**

3. Let's Analyze the Response:



Data agent responds with a text summary of the customers. In this case, it takes the values from the table and outputs 5 rows from the table as text.

We can also take this further and view the run steps by expanding the run steps in the response:



Commented [SC1]: Screenshot should fit on the page

Analyzed ecommerce-dataset SemanticModel for: "Show the top 5 rows from the Customers table, including all fields available."

Query code

Dax

```
// Return the first 5 customers (by Customer ID ascending), including all columns from the Custome
rs table.
EVALUATE
  TOPN(
    5,
    'Customers',
    'Customers'[Customer ID], ASC
  )
ORDER BY
  'Customers'[Customer ID] ASC
```

Query output

Customers Customer ID	Customers Customer Zip Code	Customers Customer City	Customers Customer State
0000366f3b9a7992bf8c76cfdf3221e2	7787	CAJAMAR	SP
0000b849f77a49e4a4ce2b2a4ca5be3f	6053	OSASCO	SP
0000f46a3911fa3c0805444483337064	88115	SAO JOSE	SC
0000f6ccb0745a6a4b88665a16c9f078	66812	BELEM	PA
0004aac84e0df4da2b147fca70cf8255	18040	SOROCABA	SP

In the response run steps you will see how the response was generated:

- User's questions *what are the top 5 rows in customers table?* was paraphrased by the agent as *Show the top 5 rows from the Customers table, including all fields available.*
- Above paraphrased query was used to generate the DAX query depending on the tables selected. You can see 'Customer' table is used in the DAX.
- Output of the executed DAX query which is used by the data agent for final response

4. Now you can try asking some additional data exploration questions. Take this time to understand the results of the query and how data agent came up with the result. For each of these questions, notice the paraphrased question, DAX and query output. Ask the following questions to start:

Commented [SP2]: Add it may be worded differently

Commented [MJ3]: Needs instructions so these should be moved

Commented [SC4R3]: These do say it's supposed to be data exploration questions, so they should technically be simple

a. *How many unique cities are sellers present in?*

b. *What is the most common product category purchased?*

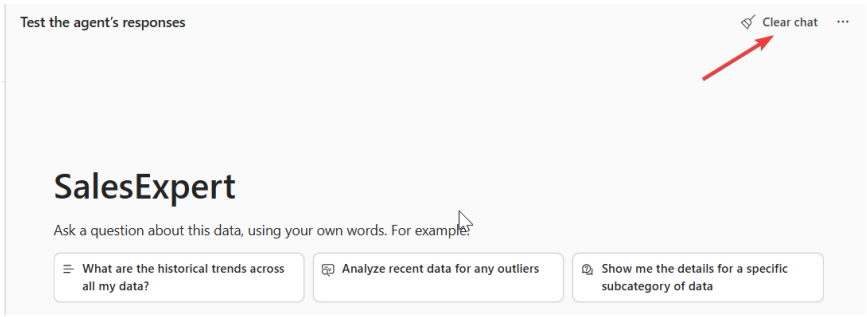
c. *What was the most expensive order purchased to date?*

Commented [MJ5]: Note what date is being considered here

Step 3: Improving Results Using Agent Instructions

The data agent consistently retains conversation history to ensure accurate contextual understanding of each inquiry. When you make any changes, clear the history for the changes to reflect.

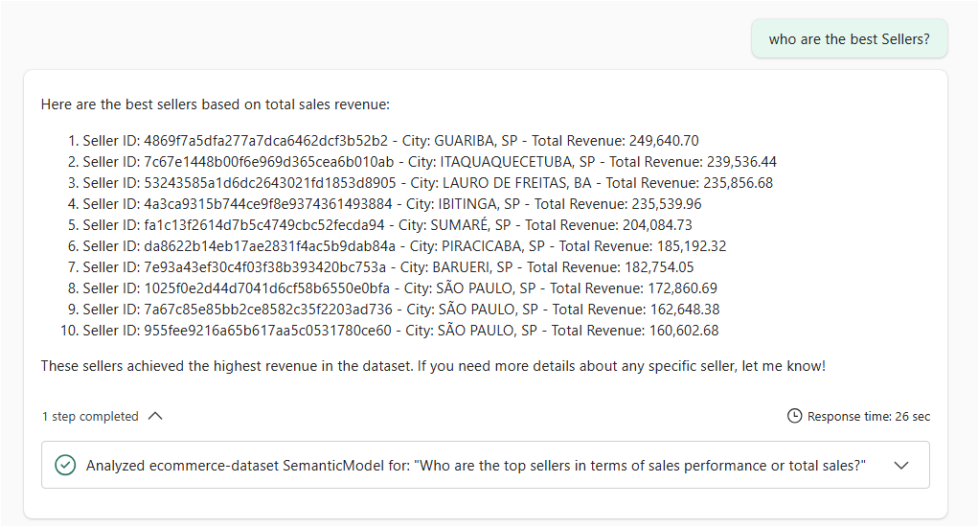
1. Let's start by clicking "Clear Chat" on the top right corner



2. Ask the agent the following question: ***Who are the best Sellers?***

Commented [SP6]: Who are the 5 best sellers?

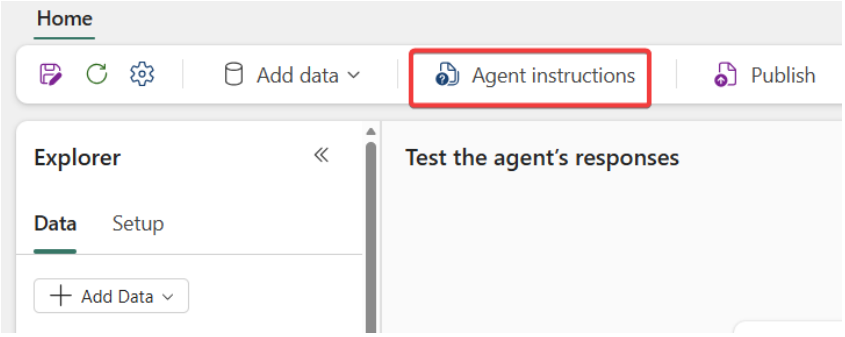
Notice the response:



The agent provides a response by analyzing all the sales data. However, you may want to define "best"

differently or want answers for a selection of the data.
We can improve and guide the agent response by providing instructions to the agent.

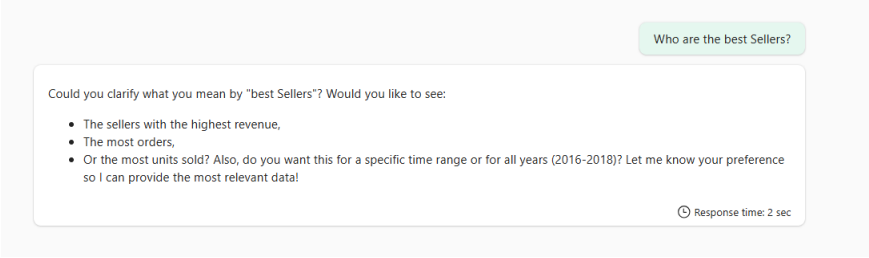
2. Select “Agent Instructions” in the Ribbon



4. Add the following instructions to the agent:

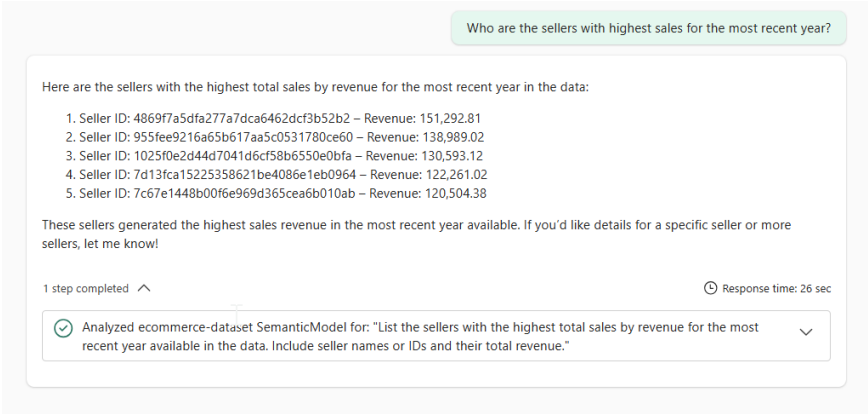
```
# Objective
You are an analytics assistant for an e-commerce marketplace. The dataset covers
orders across Brazil, including order details, products, customers, sellers,
payments, and delivery logistics.

# Guidelines
ALWAYS start by clarifying ambiguous questions first (location, metric, time
range, aggregation level, use of English labels). Example question to ask: "Do
you want top by units sold, revenue, or number of orders? Do you want the answer
for a specific location? Over what date range?"
```

5. Click “Clear Chat” on the top right corner.
6. Try asking data agent the same question again – ***Who are the best sellers?***
   Now the agent should be able to ask clarifying questions to gather additional context.



7. You can now respond with “***Who are the sellers with highest sales for the most recent year?”.***

The agent now responds clearly with relevant information:

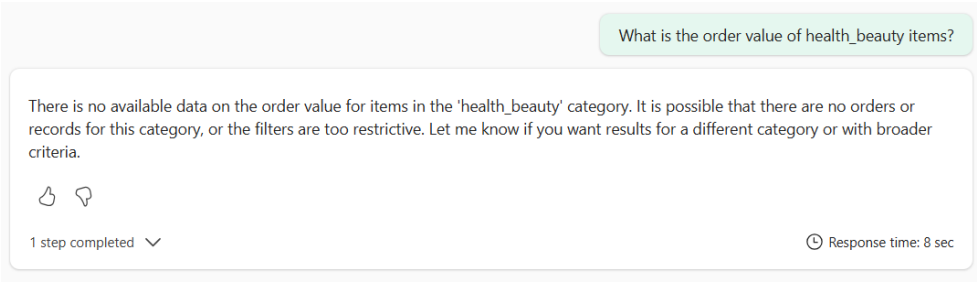


7. When you try asking data agent any of the questions below, you're likely to receive follow-ups that will allow you to explore different filters.
   (Tip: Clear chat between your questions for testing)
   a. ***Which products perform well?***
   b. ***How do order values compare between regions?***
8. As an additional exercise, try changing the agent instructions and notice how the agent responds.

Step 4: Improving Results Using Prep for AI configuration

1. Let's start by clicking "Clear Chat".
2. Ask the question: ***What is the order value of health_beauty items?***

   For the above query, you will see a response that there was no available data.



Let's analyze the response by expanding the steps:

1. First, inspect the natural language query : *What is the total order value for products in the 'health_beauty' category?* The agent orchestrator paraphrased the question correctly.
2. Then we look at the DAX query to see if correct measures, tables and columns were used based on the natural query.
    a. DAX uses [Total Revenue] measure to calculate the order value. This is correct.
    b. To filter the total revenue to only 'health_beauty' category, 'Products'[Product Category] column was used. You will also see a note *Follow instruction to use [Product Category]* in the DAX.

What is the order value of health_beauty items?

There is no available data on the order value for items in the 'health_beauty' category. It is possible that there are no orders or records for this category, or the filters are too restrictive. Let me know if you want results for a different category or with broader criteria.

1 step completed

Response time: 8 sec

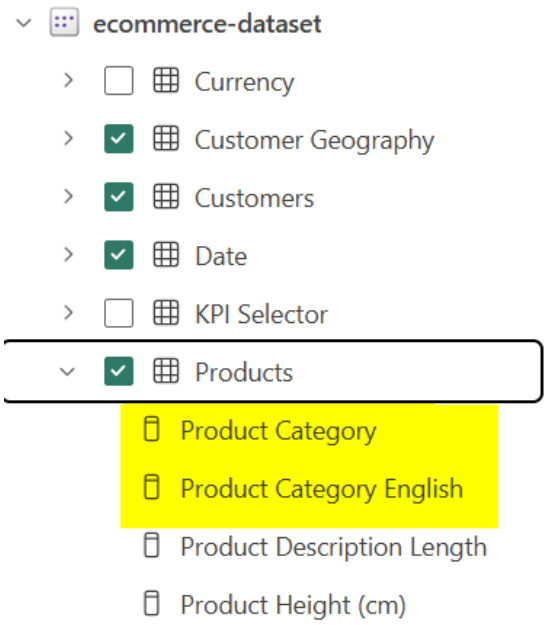
Analyzed ecommerce-dataset SemanticModel for: "What is the total order value for products in the 'health_beauty' category?"

Query code

Dax

```
// Total order value (revenue) for products in the 'health_beauty' category.
EVALUATE
  CALCULATETABLE(
    ROW(
      "Total Order Value", [Total Revenue] // Uses the core revenue measure (price + freight)
    ),
    'Products'[Product Category] == "health_beauty" // Follow instruction to use [Product Cate
gory]
  )
```

3. In the Data section, expand the Product table. You will notice that it has two columns, Product Category and Product Category English. Data agent used the Product Category column which has product descriptions in Portuguese.

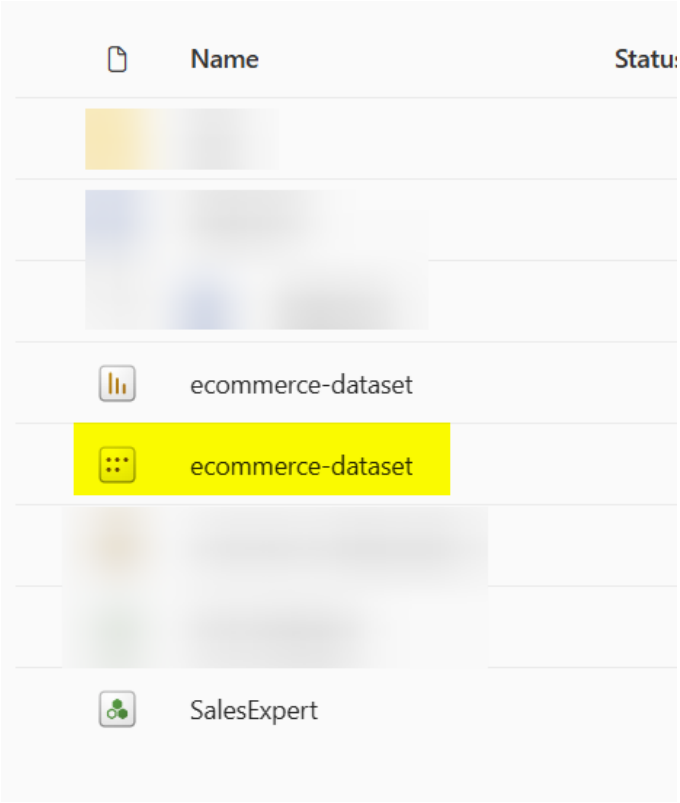


Product ID	Product Category	Product Category English
ab94356394683706626f737b7c135ae1	beleza_saude	health_beauty
848fd511a2efc8b7bc15bf948b2f363c	beleza_saude	health_beauty
be4a664850e0060b7559c67de45e8371	beleza_saude	health_beauty
39fb1b20d35ef11b96ddb1180d89c25d	beleza_saude	health_beauty
ecb6e29c8e01d5fbc73da1d0f12c088e	beleza_saude	health_beauty

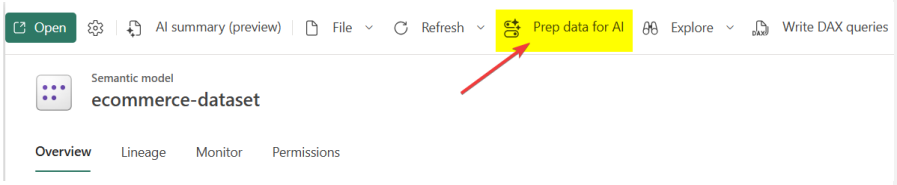
4. To fix this and instruct the data agent to use the right columns for DAX generation, we have to use 'Prep data for AI' configuration of semantic model. Agent instructions we used earlier are used for generating natural language queries, formatting the responses, routing the queries to the correct data source etc. Prep data for AI, on the other hand, influences the DAX generation.

Prep data for AI

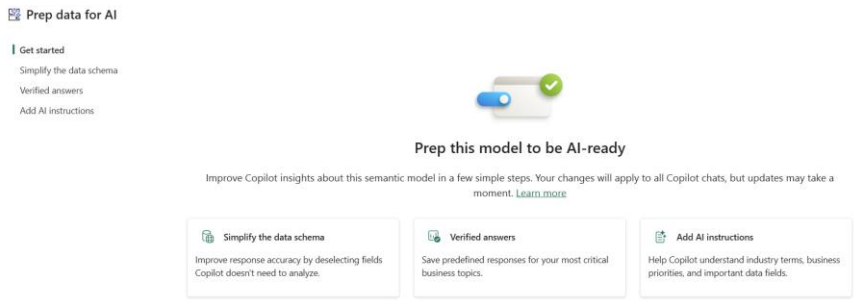
1. To open Prep data for AI configuration, browse to the workspace where the semantic model is located.
2. Open the semantic model ecommerce-dataset



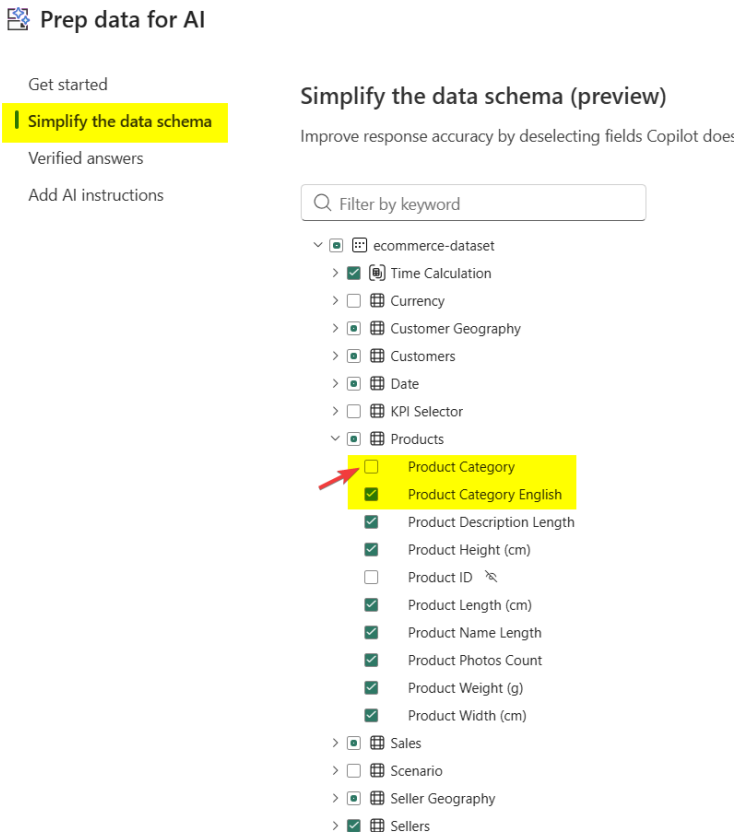
3. At the top, click on the 'Prep data for AI' option.



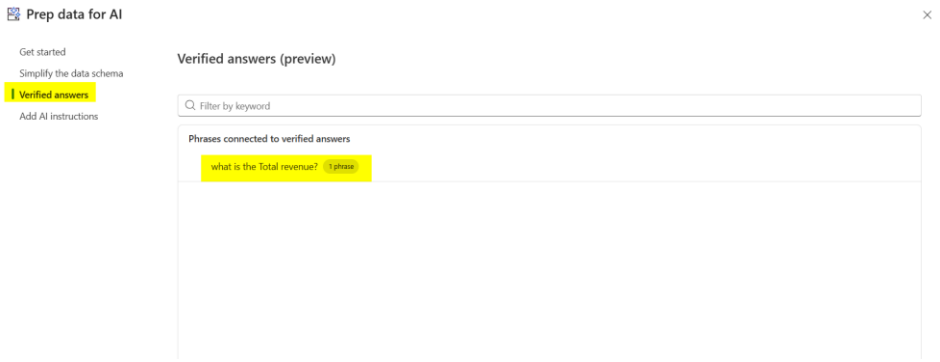
4.You will be presented with options to configure the semantic model to make it AI-ready



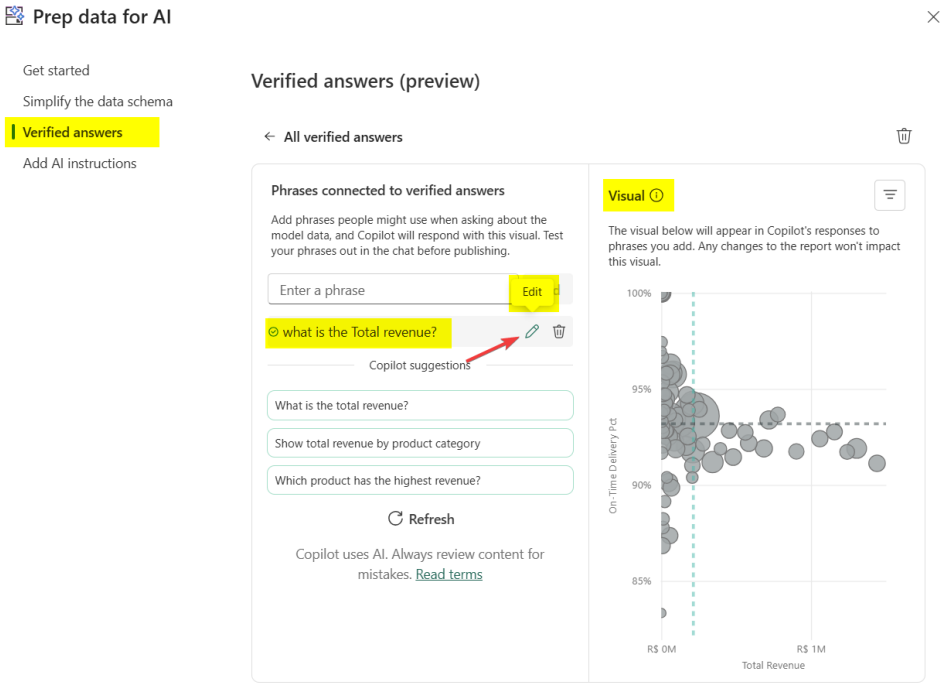
4. Select 'Simplify the data schema'. It may take a few minutes to load the schema initially.
5. Notice the tables, columns, measures and other semantic model objects that have been selected and un-selected. Only the objects that are part of the AI schema selected in this option will be considered by data agent.
6. Under Product table, un-select 'Product Category' and only keep 'Product Category English' selected. Data agent will now only consider the selected columns for DAX query generation. Click 'Apply'.



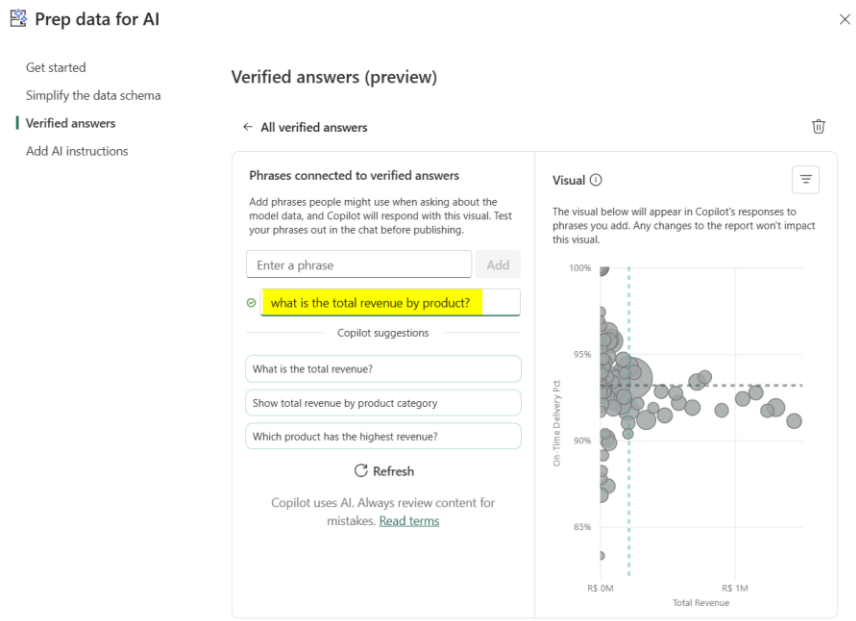
7. Next, open the `Verified answers` option.
8. Verified answers provide human-approved visual responses that are triggered by predefined phrases and questions. Each verified answer includes one or more trigger phrases, a Power BI report visual and optional associated filters.
9. To associate the 'Product Category English' column with the trigger phrase, select the already partially defined Verified answer. Select the "what is the total revenue?" phrase.



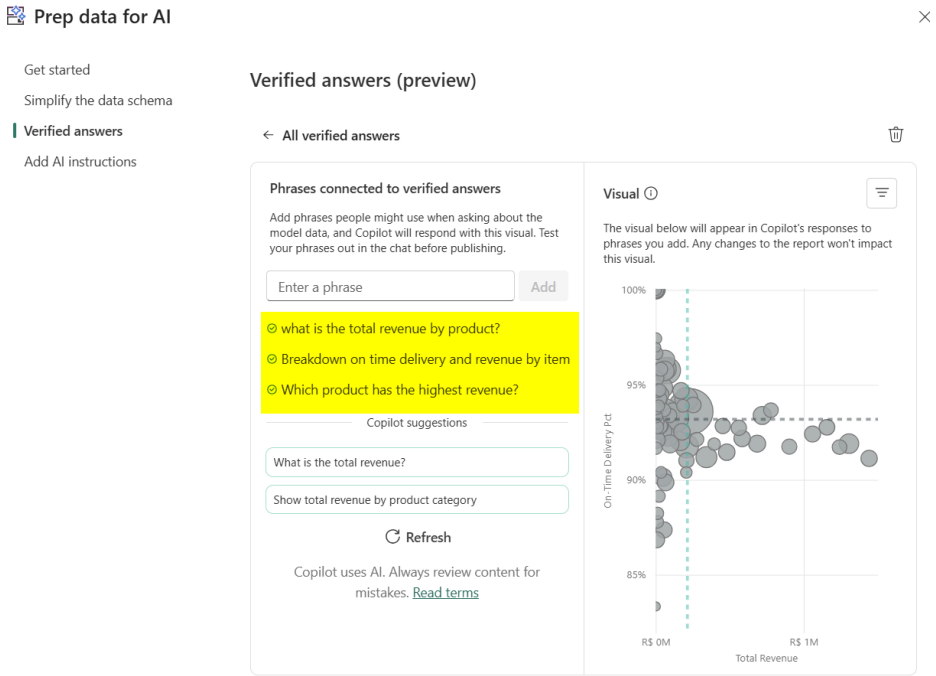
10. This scatter plot visual shows ‘On-Time Delivery pct’ vs ‘Total Revenue’ by product. It uses the ‘Product Category English’ column.



11. Select pencil option to edit the trigger phrase to: `what is the total revenue by product?`



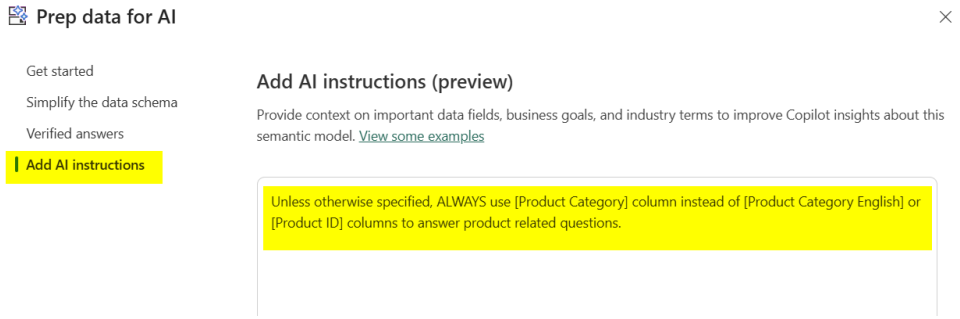
12. Add few more phrases:
    a. Breakdown on-time delivery and revenue by item
    b. Which product has the highest revenue?



13. Click 'Apply'
14. Select 'Add AI Instructions'. You should see existing instruction :

    Unless otherwise specified, ALWAYS use [Product Category] column instead of [Product Category English] or [Product ID] columns to answer product related questions.

    The instructions specifically asked the agent to use the 'Product Category'.



15. You can either remove these instructions or re-write it to: *ALWAYS use [Product Category English] column for product or item related questions.*
16. *Click 'Apply' and 'Close'.*
17. After applying the changes, wait 5-10 minutes for the changes to take effect.

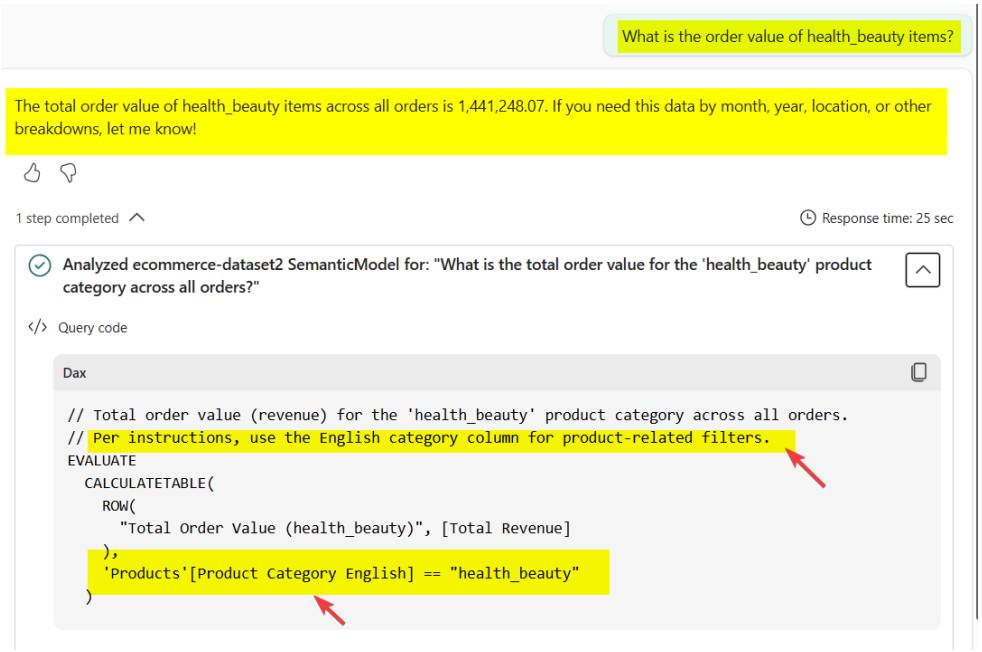
DAX generation tool now has the updated guidance on how to respond to question related to revenue and products/items.

Testing the data agent

Above, we configured the semantic model to make it AI-ready. Let's test the data agent.

1. Go to the data agent again
2. Click on 'Clear chat' to start with updated context
3. Ask the same question again: 'what is the order value of health_beauty items?
4. This time, you should see the correct response and [Product Category English] column is filtered.

   Note: Earlier we instructed the agent to ask clarifying questions in case of ambiguity. You may be asked to provide time range or additional filter context. Either answer those questions or clear the agent instructions temporarily for testing purposes.



5. Additional questions you may try :
    a. *Give me the revenue from furniture related purchases?*
    b. *How many items were purchased for kids?*
    c. *Which regions have the slowest delivery times?*

Step 5: Challenge – Use what you've learned to answer additional questions

1. From what you've learned above, can you configure the Data Agent and Prep data for AI to answer the following questions:
    a. **Who are the top 10 sellers with the most operational inefficiencies? (Considerations: Late deliveries, time to approve orders)**
    b. **Which product categories are poorly advertised? (Considerations: Product description lengths, Product photo counts.)**
    c.

Congrats, you have officially finished the lab!